Predicting Natural Disaster Economic Impact from County Demographics and Housing Conditions

Project Phase 3: Final Report

CSPC 5071 — Data Management for Data Science

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Git Repository: https://github.com/dcnguyen060899/su_cpsc5071_group_project_sad_sravya_anushka_duy

1 Problem Statement

Natural disasters have historically caused billions of dollars in economic damage across the United States. Their impact, however, varies significantly across communities, and we believe that counties with different demographic profiles, income levels, and housing conditions experience vastly different outcomes from similar events. So our central research question we are trying to answer is:

> *Can we predict the level of federal disaster assistance allocated to a county based on its pre-disaster demographic characteristics and housing cost conditions?*

Now that we had worked through all three phases of this project, our understanding of this question has been refined in important ways. Data exploration revealed that FEMA's public dataset does not include dollar assistance amounts, so our outcome variable is limited to disaster declaration counts and program activation flags. EDA further revealed that declaration frequency is heavily right-skewed and strongly concentrated in specific geographic regions. What this tells us is that modeling confirmed the most important finding: **disaster frequency is overwhelmingly driven by geography, not demographics.** State fixed effects alone explain 57.5 percentage points more variance than all six demographic and housing features combined ($R^2 = 0.167$ demographics only vs. $R^2 = 0.742$ with state indicators). Bottom line is that this shifts the interpretation of our research question because we are not predicting which counties are inherently more vulnerable, but rather which geographic regions are historically more exposed to federally declared disasters.

2 Data Sources and Preprocessing

Our pipeline integrates three federal datasets joined by standardized 5-digit county FIPS codes, implemented in two Jupyter notebooks: `initlal_data_exploratory.ipynb` (challenge discovery) and `database_pipeline.ipynb` (ETL and MySQL loading).

2.1 Data Sources

1. **OpenFEMA Disaster Declarations v2** — FY2020–2024, 16,298 declarations across 3,277 counties. Collected via paginated REST API (1,000 records per page, 19 pages total).

2. **Census ACS 5-Year Estimates, 2022** — Demographic records for 3,222 U.S. counties. Single API request; no key required.

3. **HUD Fair Market Rents, FY2025** — FMR estimates for 0–4 bedroom units across 3,228 counties, originally 4,764 rows reduced via population-weighted aggregation to resolve New England town-level duplicates.

2.2 Key Preprocessing Decisions

FIPS standardization. For each source, we encoded county identifiers differently. For example, FEMA and Census split FIPS into two separate string fields, while HUD used a single 10-digit integer. We zero-padded and concatenated to produce a uniform 5-digit string (e.g., `'41067'`). We rejected

integer conversion because it silently drops leading zeros, and that is not something we can afford in a join key.

HUD many-to-many resolution. HUD's 1,536 duplicate FIPS codes (from New England town subdivisions) would inflate disaster counts in a naive join. Think about it, without resolving this, we would be inflating disaster counts simply because of how New England jurisdictions are structured. Our solution: population-weighted averaging collapsed all town-level FMR rows into one county-level value per FIPS. This is a core data integrity decision that made the subsequent analysis trustworthy.

Census sentinel replacement. The Census API uses −666,666,666 for suppressed data. One county had this value in `median_household_income`; we replaced it with `NULL` before any downstream analysis. Imagine if we had not caught this, that would have been a catastrophically wrong income value feeding directly into our models.

2.3 Database Design

The cleaned data was loaded into a MySQL relational database (`disaster_impact_db`) implementing a star schema with four tables. The `county` dimension table (3,292 rows) uses `FIPS CHAR(5)` as the primary key such that the union of all FIPS codes across all three sources. The `disasters` fact table (16,298 rows) stores one row per declaration event with a foreign key to `county`. The `demographics` and `housing_costs` dimension tables each have one row per county (partial coverage). Two SQL views were created: `county_disaster_summary` (one row per county) and `analysis_ready` (full `LEFT JOIN` across all four tables, 3,292 rows × 17 columns), which served as the starting point for all EDA and modeling. We chose the `LEFT JOIN` design intentionally for information preservation because the fact that, without it, those 80 missing counties would have silently disappeared instead of being explicitly acknowledged.

3 Exploratory Data Analysis

All EDA was performed in `eda.ipynb` on the `analysis_ready` view. We highlight the findings that most directly shaped our modeling decisions, and we want to be upfront that EDA is what raised the questions we then used modeling to answer.

3.1 Target Variable Distribution

As we diagnose in EDA thoroughly, `total_disasters` is heavily right-skewed (mean 5.7, median 4, max 149), with most counties clustering at low counts and a long tail of high-frequency counties. So what do we do for modeling to be interpretable? We log transform them such that it reduces the right skew. A log transformation $\log(1 + \texttt{total_disasters})$ produces a near-symmetric distribution suitable for OLS regression. For classification, we defined a binary `high_impact` flag as counties above the median (more than 4 declarations), producing a near-balanced split (48.9% high-impact). Bottom line is that our models can be evaluated straightforwardly without any class imbalance corrections.

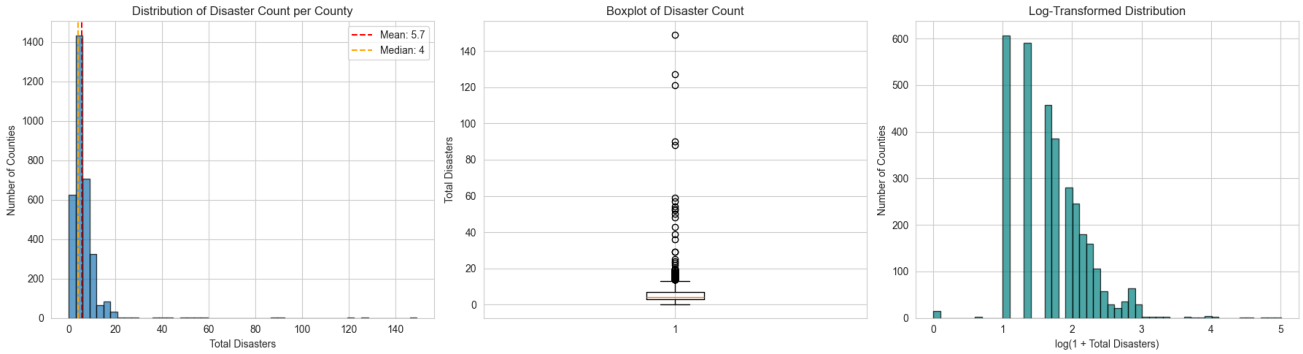

Figure 1: Distribution of `total_disasters`: raw histogram (left), boxplot (center), and log-transformed (right).

3.2 Feature Engineering

Raw demographic counts are not comparable across counties of different sizes, and this is why we need to be thoughtful here. We derived three normalized features: `poverty_rate` = poverty count / total population × 100 (mean 14.6%, max 65.6%), `unemployment_rate` = unemployment count / labor force × 100 (mean 5.2%), and `log_population` = $\log(1 + \texttt{total_population})$ to compress population's 50-to-10-million range. All five FMR columns are correlated above 0.94 with each other; we selected `fmr_2` (2-bedroom Fair Market Rent) as the single housing cost representative, following HUD's own affordability benchmark convention. Using all five would just introduce collinearity without adding new information.

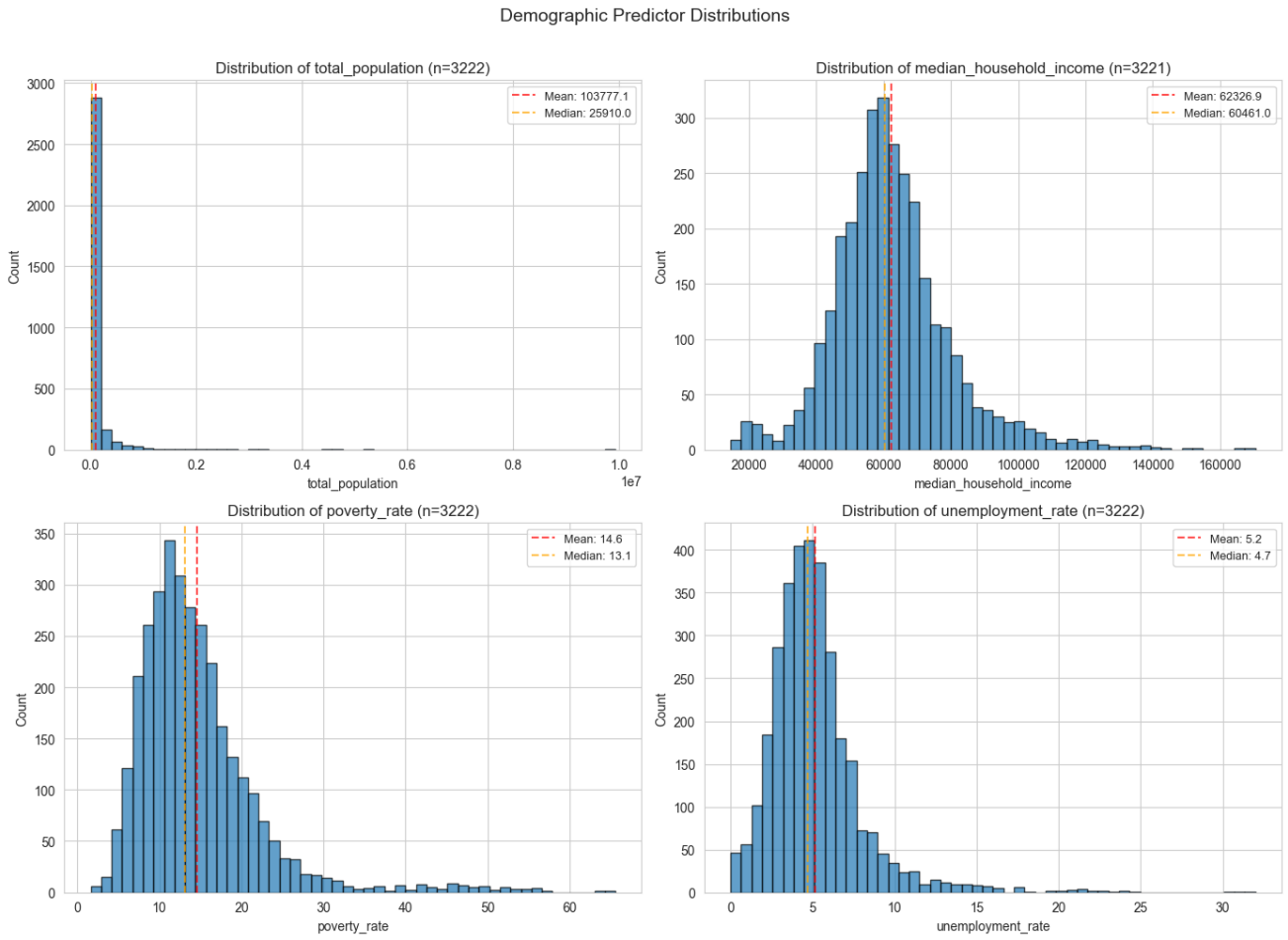

Figure 2: Distribution of demographic predictor variables across 3,222 U.S. counties.

3.3 Geographic Concentration and the Key Limitation

State-level aggregation revealed that Maine leads with approximately 40 mean declarations per county, followed by Gulf Coast states (Louisiana, Florida) and tornado-alley states (Kentucky, South Carolina). The bottom ten states average fewer than 3 declarations per county. This geographic concentration highlighted the central challenge of our feature set, and we want to be honest about this: **a county's position on the Gulf Coast hurricane track or in tornado alley is not encoded in its poverty rate, median income, or housing cost.** We designed the modeling phase specifically to test whether demographics retain any predictive power once geography is controlled for. This is a question we were already suspecting from EDA, but we needed the models to actually prove it.

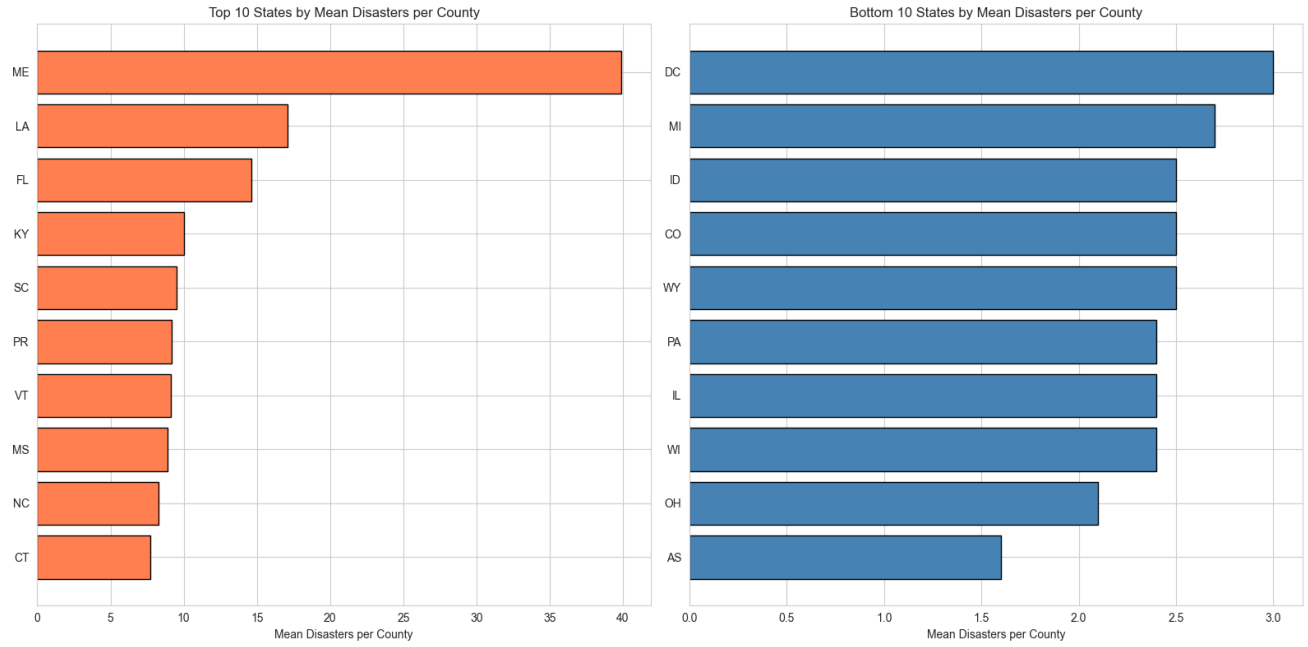

Figure 3: Top 10 and bottom 10 states by mean disaster declarations per county.

3.4 Bivariate Correlations

Among substantive predictors, `poverty_rate` shows the strongest positive Pearson correlation with `total_disasters` ($r = +0.218$), followed by `unemployment_rate` ($r = +0.173$). `median_household_income` shows a weak negative association ($r = -0.126$). As you can see, all six scatter plots display wide scatter with near-flat linear trends, and this further tells us that demographic features alone are weak predictors of disaster frequency. We are not surprised by this, just with a simple fact that we anticipated it from the geographic concentration finding above.

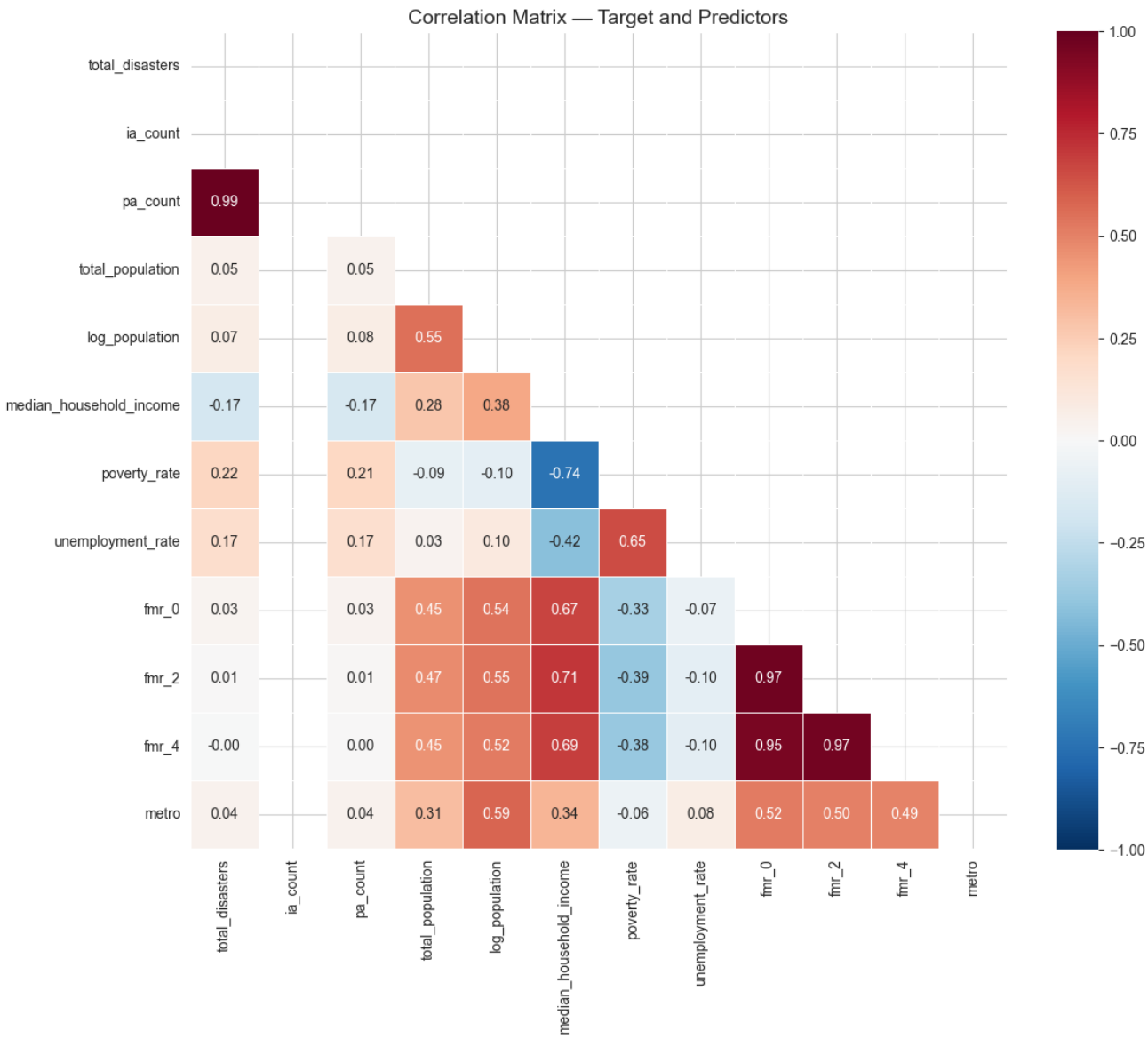

Figure 4: Bivariate scatter plots of each predictor against `total_disasters` with OLS trend lines.

4 Modeling

Now that we had finished building our database pipeline and completed our exploratory data analysis, this section will walk us through the statistical models we built to answer the questions our EDA raised. For this modeling study, our aim is to interpret semi-complex models, and by what we mean by semi is that they are not black boxes like deep learning. Our plan is to build three models: OLS regression, logistic regression, and a decision tree. We chose these not for maximum predictive power but for transparency and pedagogical value. We anticipated that each model would teach us something different about the data.

As we diagnosed in EDA thoroughly, there are still 80 missing rows despite after we left joined with our clean database design (for information preservation). We need our "cleaned" database to be even more clean for modeling to work because these models can't process missing values. After dropping the 80 counties with any NULL value (2.4% of rows), the complete-case dataset contains 3,212 counties. We

used an 80/20 stratified train/test split (`random_state=42`), with `StandardScaler` fit on training data only to prevent data leakage.

One thing to notice for the split: remember that we have a 48.9 percent high-impact class ratio. When splitting into train and test sets, we ensured stratification equally; because without it, random chance could put a disproportionate number of high-impact counties in one split, and that would make our evaluation misleading, basically unreliable. This is a sampling technique protocol we learned from statistical and experimental design course. Cool, the 80/20 train/test split showed us 2,569 training and 643 test counties, with the high-impact class proportion nearly identical in both sets (48.9 percent and 48.9 percent), confirming that stratification worked as intended.

Also, from EDA, we know that our `median_household_income` is in tens of thousands. What this means is that if we don't scale this, our modeling would learn some proportion of the data to be numerically dominant over other features like `metro` (binary 0/1) or `poverty_rate` (0–65). Therefore, we use `StandardScaler` to put all features on the same zero-mean, unit-variance scale so that the model evaluates each feature on equal footing.

4.1 Model 1: OLS Linear Regression

We used `statsmodels` OLS (not scikit-learn) because it provides full statistical inference, particularly p-values, confidence intervals, F-statistic, and R^2; which are essential for a data management course focused on interpretation rather than raw predictive accuracy. We believe that this is a data management course for data science and not a statistical machine learning course, therefore the goal is to demonstrate that a well-designed data pipeline produces a dataset that supports meaningful statistical analysis.

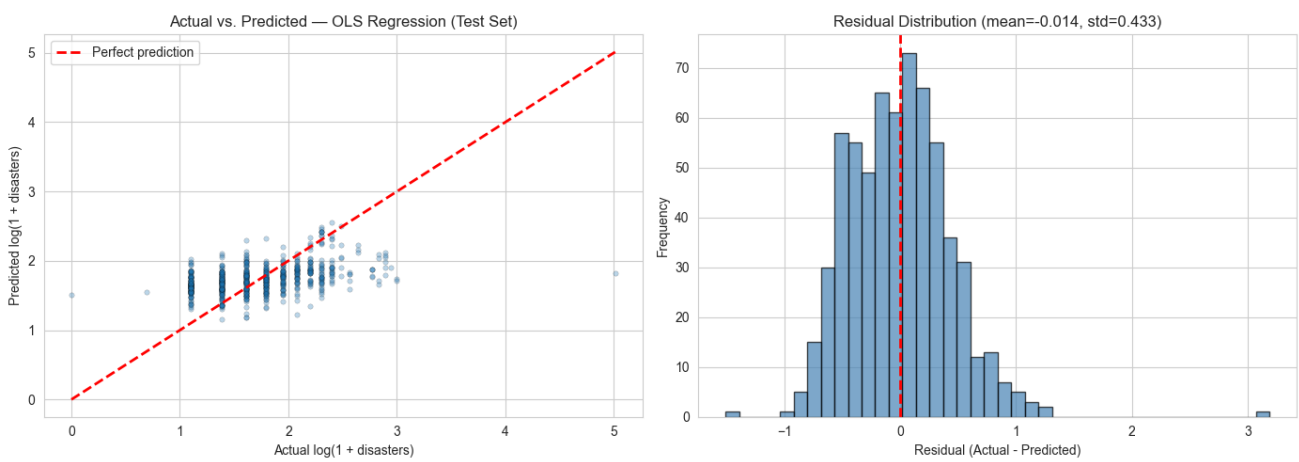

Figure 5: Actual vs. predicted $\log(1 + \text{disasters})$ (left) and residual distribution (right) for OLS on the test set.

Not to our surprise, the OLS model explains only 16.2 percent of the variance in log-disaster counts ($R^2 = 0.162$ on the training set and $R^2 = 0.185$ on the test set). This confirmed us the fact that demographics alone are weak predictors of disaster frequency. The F-statistic ($F = 82.57$, $p < 10^{-90}$) confirms the model as a whole is statistically significant. At the individual feature level, four of six predictors are significant at $p < 0.001$, keep in mind, however, these are coefficients not correlation ρ, so they are partial effects:

Feature	Coef	Std Err	P> $\|t\|$	Direction
`log_population`	0.042	0.012	0.001	↑ More disasters
`median_household_income`	−0.151	0.019	< 0.001	↓ Fewer disasters
`poverty_rate`	0.102	0.017	< 0.001	↑ More disasters
`unemployment_rate`	0.009	0.012	0.488	— Not significant
`fmr_2`	0.133	0.015	< 0.001	↑ More disasters
`metro`	0.011	0.012	0.340	— Not significant

We see that `fmr_2` (which is the fair market rent) and poverty rate have a positive coefficient, increasing the disaster likelihood, while higher income reduces it, and unemployment and metro status are not statistically significant. The model predictions, as you can see, cluster in a narrow band around 1.5 to 2.0 regardless of the actual value (remember this is log), implying that it fails to capture high disaster counties. The right-skewed residuals confirm us that it systematically underpredicts for most disaster-prone counties. The residuals are roughly bell-shaped and centered at zero, though with heavier tails than a normal distribution (Jarque-Bera test rejects normality), which is expected given the skewed nature of disaster counts. This is consistent with our finding that geography, not demographics, drives extreme disaster counts.

4.2 Model 2: Logistic Regression

Now that we see OLS showed that demographics can't precisely explain disaster counts as we observe in $R^2 = 0.162$, this means that 84 percent of the variance is unexplained by the model. However, note that it is not completely useless. Tt just means that we should ask a simpler, more achievable question: can we at least classify which counties are more disaster-prone than others instead of putting our full faith demographically? So naturally, we would think of logistic regression, such that we can reframe the problem from predicting exactly how many disasters to instead asking whether a county is above or below the median. This binary yes and no allowed us to be honest with the demographic being a weak signal. So the move from regression to classification is a pedagogical decision to lower our ambition to match what we believe the data can actually support!

For the binary classification task (`high_impact`: above-median disaster count), we fitted a logistic regression on the scaled feature set.

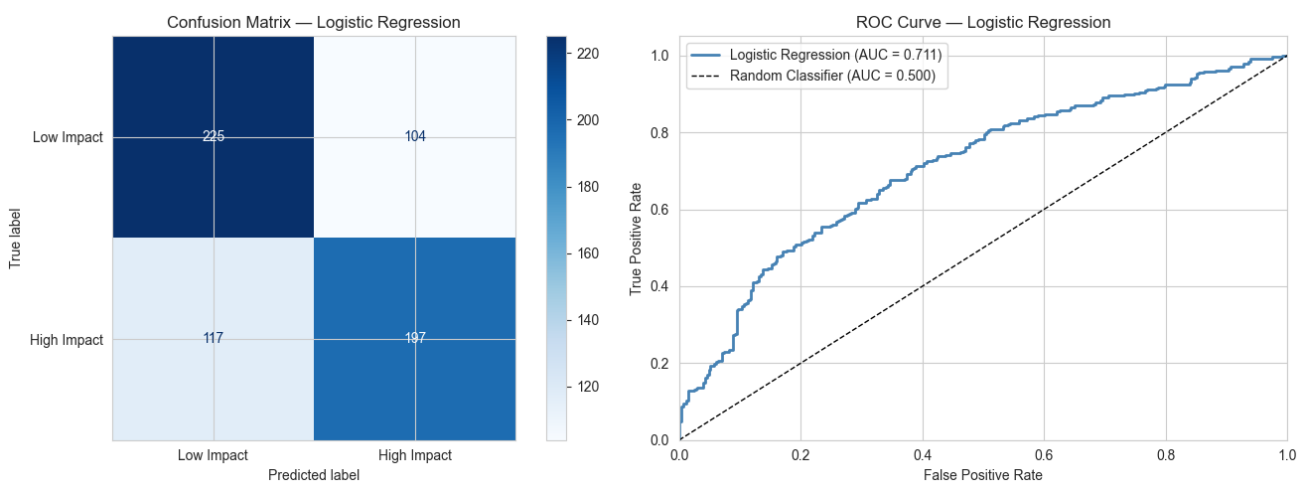

Figure 6: Confusion matrix (left) and ROC curve (right) for logistic regression on the test set.

From both the confusion matrix and the visualization, we can observe that our logistic regression

correctly classifies 66 percent of counties and achieves an AUC of 0.711. Now compare this result to OLS, we can clearly see that this is meaningfully better than random chance, but the 117 missed high-impact counties (false negatives) and 104 false alarms suggest that our demographic features carry real but limited signal, again telling us that geography remains the missing piece. The AUC meaning is that the logistic regression model is asking: across all pairs of one high-impact and one low-impact county, does the high-impact county get a higher probability score of being classified as high-impact? As we see, 71.1 percent of the time it does.

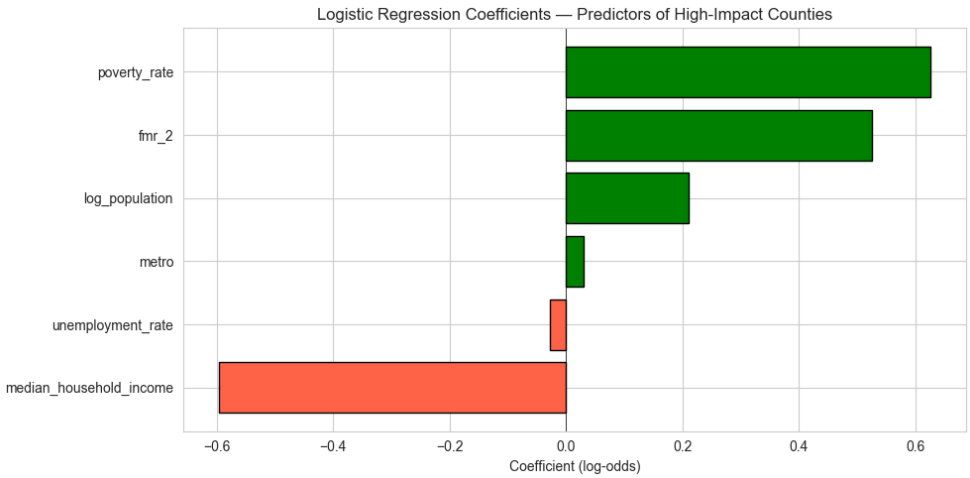

Figure 7: Logistic regression coefficients (log-odds scale). Features are standardized, so magnitudes are comparable.

Now the model perfectly tells us how well the model classifies. However, not why it is but rather so that it is still a black box performance number in terms of its parameterization. So now we want to know which features are actually driving this prediction capacity. The coefficients tell us the direction and magnitude of each feature's contribution to the model's final output. Because the features have been standardized, a one standard deviation increase in that feature changes the log odds of being high-impact by the amount of that coefficient.

The coefficient bar chart confirms the pattern from OLS: `poverty_rate` (+0.626) and `fmr_2` (+0.526) are the strongest positive predictors, while `median_household_income` (−0.597) is the strongest negative predictor. Same story (coin), different side of the coin from OLS that `poverty_rate`, `fmr_2`, and `median_household_income` were the only statistically significant features, and unemployment and metro were not significant, as you see a small contribution to log odds in terms of coefficient magnitude.

4.3 Model 3: Decision Tree Classifier

Now that we see logistic regression told us what predicts disaster likelihood and how confidently we are (probability wise), note that the coefficients are in log odds. So we need statistical literacy to interpret them. Let's think about decision tree. We want to ask, can we express the same findings as a simple flowchart that anyone can follow from the top down? For example, if poverty rate is above X and income is below Y, classify as high-impact without the audience needing to know log-odds at all. So keep this in mind: we are moving from statistical inference to a more human-readable decision tree, trading off some of the mathematical rigor for interpretability. Paradoxically enough, decision tree can let us capture non-linear relationships and feature interactions that logistic regression's linear boundary might miss.

We fitted a constrained decision tree (`max_depth=4`, `min_samples_leaf=50`) on *unscaled* features so that split thresholds remain in their original units and are directly interpretable.

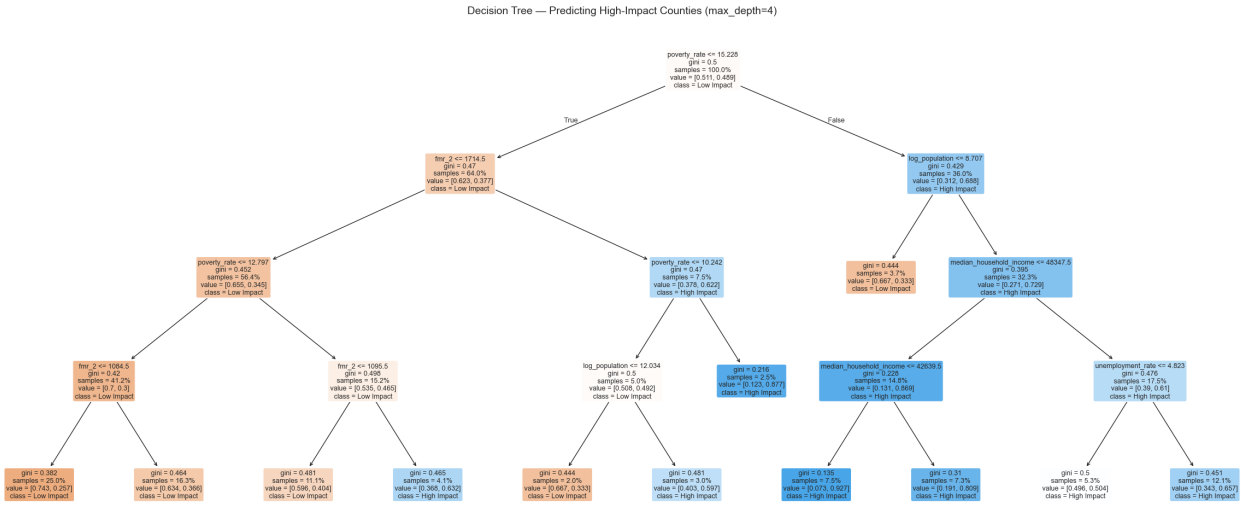

Figure 8: Decision tree visualization (depth 4). Each node shows the split rule, Gini impurity, and class proportions.

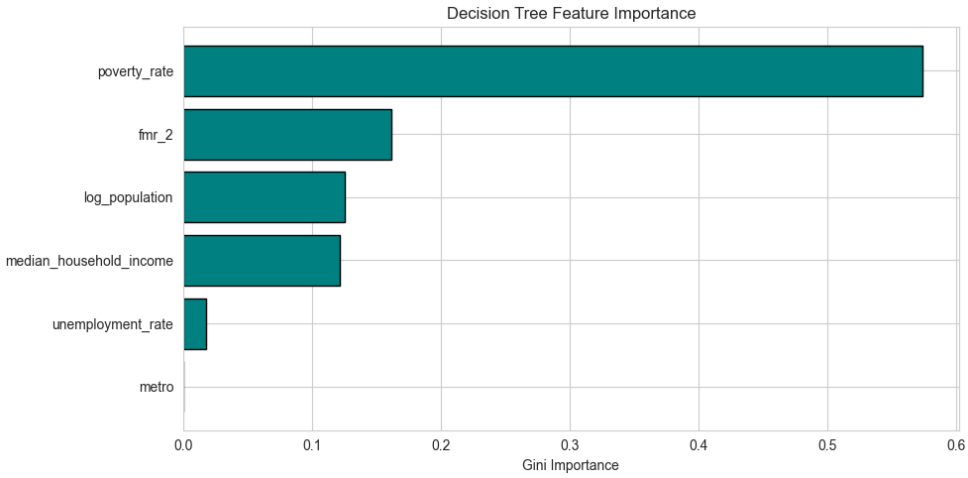

Figure 9: Gini feature importance from the decision tree. `poverty_rate` accounts for 57% of all splits.

Looking at the confusion matrix table and the tree itself, notice that the same pattern converges to the same story with OLS and logistic regression, poverty rate is the first and most important split at the root, and counties with high poverty, high fair market rent, and low income consistently branch toward high impact. Well, because these three features capture the same underlying reality. Let's step back and think about this intuitively: economically vulnerable counties (poor, expensive housing, and low income) tend to cluster in geographically high-risk regions like the Gulf Coast and tornado alley, so the signal is real and consistent regardless of which model or method is being used to detect it.

Though the tree above is a bit hard to visualize with all the rules, we want to introduce Gini importance such that it collapses the entire tree into a single ranked bar chart, which gives the same features that drive the outcome that the logistic regression coefficients had told us, but now from the tree perspective. Just a quick note on Gini importance meaning: at each node, if we split a node 50/50, the node is 0.5 Gini, meaning it is maximally impure, and a node with all one class is perfectly pure. Gini importance

for a feature is simply the total impurity reduction that feature is responsible for across all splits in the tree.

The decision tree achieves accuracy 0.656 and AUC-ROC 0.700, virtually identical to logistic regression. The Gini importance ranking reveals that `poverty_rate` alone drives 57.3% of the tree's splitting decisions. What does this tell us? This means that poverty as a feature alone is responsible for over half of all impurity reduction in the tree. The `metro` flag contributes zero importance, meaning the tree never finds a useful split on urban/rural status once the other features are accounted for. But bottom line is that this perfectly mirrors the same story with OLS and logistic regression.

4.4 Model Comparison

Model	**Task**	**Accuracy**	**Precision**	**Recall**	**AUC**
Logistic Regression	Classification	0.656	0.654	0.627	0.711
Decision Tree (d=4)	Classification	0.656	0.663	0.602	0.700
OLS (demographics)	Regression		$R^2 = 0.167$		—
OLS + State FE	Regression		$R^2 = 0.742$		—

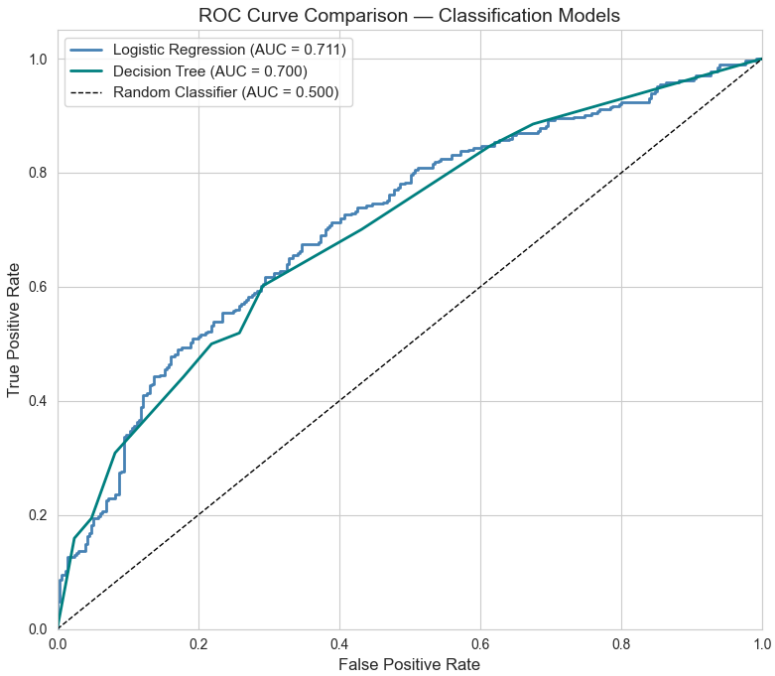

Figure 10: Overlaid ROC curves for logistic regression (AUC 0.711) and decision tree (AUC 0.700).

Nothing surprised here, as we thoroughly and rigorously analyzed previously, both models perform nearly identically. This tells us there is some ceiling fundamentally from our data itself (missing geography features), not the choice of modeling. So switching to a more complex algorithm won't help until we add geographic features. The near-identical accuracy between a linear model (logistic regression) and a non-linear model (decision tree) further implies that the relationship between demographics and disaster classification is primarily linear and that non-linear interactions among these six features add little predictive power.

4.5 State Fixed Effects: Geography vs. Demographics

This is the most important analysis in the project, and not to our surprise, we actually expected this the whole time. Adding 50 state dummy variables to the OLS model raises R^2 from 0.167 to 0.742, that

is a jump of +0.575.

Model	R^2	Adj. R^2	AIC
Demographics only	0.167	0.165	4,035
Demographics + State FE	0.742	0.737	369

That is a 57.5 percentage point jump. This definitely confirmed us that geography, not demographics, is the dominant driver of disaster frequency. Not that we are saying it is completely useless to include demographic features, but in fact our six demographic features were only explaining the leftover within-state variation all along. The AIC drop from 4,035 to 369 confirms the state-FE model is vastly better-fitting. That is the smoking gun.

More revealing is how demographic coefficients change when state fixed effects are added:

Feature	**Demo-only sig.**	**With State FE sig.**	**Interpretation**
`log_population`	***	***	Robust — genuine within-state effect
`median_household_income`	***	*	Weakens but survives
`poverty_rate`	***	ns	Disappears — geographic confound
`unemployment_rate`	ns	ns	Never significant
`fmr_2`	***	ns	Disappears — geographic confound
`metro`	ns	*	Flips sign (negative within-state)

*** $p < 0.001$, * $p < 0.05$, ns = not significant.

So after adding 50 columns into modeling, turned out `poverty_rate` and `fmr_2`, our two previous strongest demographic predictors, now became statistically insignificant the moment we added state in to controlled for. What we mean by controlled for is that as we added state dummies, we are holding geography constant, and demographic and housing features became insignificant in modeling. We can confirmed that they are confounded by geography all along that poor, high-rent counties aren't disaster-prone because of their demographics, they just happen to be located in high-disaster states. `poverty_rate` collapses from highly significant ($p < 0.001$) to completely non-significant ($p = 0.82$) once state is controlled for. Same happens to `fmr_2`. The only feature that survives with consistent significance is `log_population`, which remains significant ($p < 0.001$) in both models with nearly identical coefficients (0.0273 vs. 0.0266). This is the one genuine demographic predictor: larger counties attract more federal declarations regardless of state.

5 Conclusions and Limitations

5.1 Key Findings

Let's recap what we found:

1. Disasters are a geographic phenomenon, not a socioeconomic one. The state fixed effects analysis answers our research question definitively: geography explains 3.5× more variance in disaster frequency than demographics. Knowing which state a county is in is far more predictive than knowing its poverty rate, income, housing cost, or unemployment rate. This is the headline finding that demographic alone are weak predictors, but once geography is controlled for the model explains 74 percent of variance.

2. Poverty's association with disasters is a geographic confound. This is the most policy-relevant finding, and we think about it as the smoking gun of this entire study. Without state controls, poverty rate appears to strongly predict more disasters ($p < 0.001$, $r = +0.218$). A naive interpretation would conclude that poverty increases disaster vulnerability. The fixed effects analysis shows this conclusion is wrong because poverty and disaster frequency co-occur because they share a geographic driver (Gulf Coast and Deep South counties are both poorer and more disaster-prone), not because

one causes the other. Any study of disaster vulnerability that does not control for geography risks this confound. In other words, poor counties aren't disaster-prone because of their demographics, they just happen to co-locate in the same high-risk geographies.

3. Population size is the only robust demographic predictor. `log_population` remains highly significant ($p < 0.001$) before and after controlling for geography, with a stable coefficient (≈ 0.027). Larger counties cross the federal declaration threshold more often because they have more infrastructure, more people, and more area exposed to any given hazard.

4. The data pipeline directly enabled this conclusion. The fixed effects analysis required reliable state identifiers on every county row. That was only possible because our `county` dimension table established an authoritative FIPS registry and the `LEFT JOIN` design preserved all counties including those with missing demographic data. Imagine without the `LEFT JOIN` design, those 80 missing counties would have silently disappeared instead of being explicitly acknowledged. A poorly designed pipeline with mismatched FIPS keys or silent INNER JOINs would have produced biased coefficients and potentially missed the confound entirely. Those we need to include special thanks to our group work on database engineering pipeline that made all of this possible!

5.2 Limitations

There are some limitations we want to acknowledge openly:

Observational data. Despite in this story, we found out that geography was confounding demographic and housing data, we can't still clearly say that geography causes. Tt is more like correlations. We cannot establish causal relationships. The association between population size and declarations may reflect administrative factors (larger counties have more resources to apply for declarations) rather than underlying vulnerability.

Cross-sectional snapshot. We also only look at a single snapshot in time. All data reflects a single point in time, and disaster patterns and county demographics change; the relationships we observe may not hold in future periods.

Declaration counts, not damage. FEMA's public dataset measures the *frequency* of federally declared disasters, not their severity or economic cost. A county with 20 minor declarations is counted the same as a county that experienced one catastrophic event. Our target variable measures administrative activity, not true vulnerability.

Dropped counties. Remember the 80 counties that were dropped? These are systematically the most disaster-prone countie. For example, territories and frontier geographies like Alaska, that our models don't speak to. The 80 counties excluded by listwise deletion have a higher mean disaster count (10.8 vs. 5.4), exclusively Public Assistance activations with no Individual Assistance, and geographically confined to US territories, Alaska non-standard jurisdictions, and Connecticut's restructured county-equivalents. Model conclusions do not generalize to these geographies.

Six features is a limited set. Only six features were used to study disaster outcome. Intuitively, we actually need more features like elevation, distance to coastline, proximity to major rivers, historical climate patterns, and local building codes. These would move the needle but were outside our database scope.

5.3 Implications for Data Management

In short, this project demonstrates that modelling without database designing and pipelining are only as trustworthy as the data underneath what structures them, therefore, starting with a well-designed database is a must before even thinking about analysis, let alone modeling. Every modeling decision in this project, whether feature engineering, listwise deletion accounting, state fixed effects, it all relied on the integrity guarantees established at the database layer: primary key constraints preventing duplicate FIPS entries, foreign key constraints ensuring referential integrity across tables, and the `LEFT JOIN`

design preserving all counties regardless of source coverage. The finding that poverty is a geographic confound is only credible because we can verify that our join keys are correct and our coverage gaps are fully documented. So every model here ran on the `analysis_ready` VIEW our ETL pipeline built. Without referential integrity on FIPS codes, our state fixed effects would be contaminated. Bottom line is that garbage in, garbage out applies equally to causal conclusions as it does to predictions.

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